

Samanpreet Kaur

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SKILLS

Programming: C++, Java, Python, JavaScript, HTML, CSS, SQL, React, ExpressJS, Kotlin, DSA, Cloud Computing

Database: DynamoDB, SQL (MS SQL Server), PostgreSQL, Firestore

Tools: AWS (EC2, S3, VPC, IAM), Git, IntelliJ IDEA, Visual Studio Code, Android Studio, MATLAB, Microsoft Office skills

Soft Skills: Teamwork, problem-solving, eye for detail, confident, leadership, strong communication, public speaking

PROFESSIONAL EXPERIENCE

Teaching Assistant, Simon Fraser University, Burnaby, BC | DSA, Research papers, MS teams, Excel August 2025 -current

Data Structures and Programming; Social Issues and Communication Strategies in Computing Science CMPT 105W

- Facilitate **weekly** one-on-one **office hours** to provide instructional support, **exam grading**, and constructive feedback, resulting in measurable improvement in student writing and technical communication skills.
- Translate **complex technical concepts** into **clear, student-friendly explanations**, helping solve coding problems, fostering academic confidence which help them deepen DSA knowledge and frame their essays well.
- Collaborate** with **faculty** and fellow TAs to coordinate instruction, manage grading timelines, and demonstrating effective **teamwork** which ensures delivery of necessary feedback timely on MS teams.

BUSINESS ANALYST PROJECT

Ferry Reservation System | C++, GitHub, Binary File I/O, Requirement Analysis, Kanban

May 2025- August 2025

- Led end-to-end **requirements gathering** by collaborating with professor (acted as the business) over the course of 3 weeks to fetch the business needs through conversations, define **functional and non-functional requirements**, map user workflows, and translate ambiguous expectations into structured specifications.
- Prepared all **software engineering documents** from scratch including requirements analysis, UI, architectural and detailed design document with deep emphasis on scrutiny, features and design tradeoffs.
- Performed **data modelling** and built a **C++ application** to manage ferry reservations, vehicle check-ins by applying **agile practice**, closely following Software Development Life Cycle (**SDLC**) and real-world system design.
- Coordinated with team using the version-control on **GitHub** and using Process management tools like **Kanban system** to maintain consistency which promoted workflow transparency.

CERTIFICATIONS

Entry Certificate in Business Analysis (ECBA), *IIBA (International Institute of Business Analysis)*

February 2026 – current

- To gain deeper insight about requirements gathering, stakeholder analysis, business process modelling and BA fundamentals

SOFTWARE PROJECTS

Backstage Concert Companion Mobile Application | Kotlin, Android Studio, Firebase, MVVM

October 2025- December 2025

- Developing a **concert-tracking application with an intuitive multi-tab interface**, enabling users to explore events, connect with friends, and manage interests through a smooth, modern UI **using MVVM architecture**.
- Implementing **real-time event search by integrating Ticketmaster's API**, allowing users to browse local and international concerts with live updates and personalized content based on location and preferences.
- Building a scalable user account system **using Firebase Authentication and Firestore**, supporting secure login, profile management, saved events, and synced cross-device data for a consistent user experience.

Mobile Workout Tracking Application | Kotlin, Android Studio, SQLite

September 2025- December 2025

- Designing an Android fitness application with **interactive UI enabling** users to record, view, and manage workouts.
- Integrating 3 workout modes: manual input, GPS-based tracking using Google Maps API, and automatic activity inference leveraging accelerometer sensor data, using **machine learning classifier** and **Kotlin**.
- Built a user profile module with camera/gallery integration for profile images, privacy preferences, and unit selection.
- Optimizing data persistence and retrieval by implementing **SQLite database management** and **Android lifecycle handling** for reliable performance.

Squat Posture Classifier Machine Learning | Python, MediaPipe, scikit-learn

May 2025- July 2025

- Implemented a KNN-based classifier to detect correct vs. incorrect squat posture using a video input from client using python, scikit-learn.
- Optimized accuracy through feature engineering and model tuning to achieve 80% accuracy.
- Integrated MediaPipe pose estimation to extract key landmarks, enabling reliable real-time feature extraction.

EDUCATION

Bachelor of Computer Science; concentration in artificial intelligence (AI)

Simon Fraser University, Burnaby, BC

January 2025 - **August 2027** (Expected Graduation)

Associate of Computer Science

Langara College

August 2022-December 2024

PUBLIC INTERACTION EXPERIENCE

Coding Tutor / Mentor, Women in Computer Science (WiCS), SFU

January 2026– current

- Mentor WiCS students in algorithmic problem solving by teaching LeetCode-style coding problems and guiding them through efficient solution strategies in 1.5-hour study sessions.
- Foster collaborative technical learning sessions by breaking down data structures and algorithm concepts to help peers improve coding interview readiness.

Hack-The-Sem Ambassador, Google Developer Student Club, SFU

January 2026– current

- Promote the Hack-The-Sem innovation program by encouraging student participation and sharing opportunities across personal academic and professional networks.
- Contribute to program visibility and engagement by supporting outreach initiatives that attract students interested in building impactful technical projects.

Vancouver Police Associate, Kitsilano Fairview Community Policing Centre, Vancouver, BC

August 2023 – May 2025

- Resolved 15+ resident phone queries per day by directing callers to appropriate police or city services, leading to residents feel more aware and improve their access to community resources significantly.
- Partnered with Vancouver Traffic Police and ICBC to check the speed of 100+ vehicles per shift using radar equipment, assisting in enforcement actions to halt speeding.

LEADERSHIP EXPERIENCE

Student Ambassador, Simon Fraser University, Burnaby, BC

January 2026 – Present

- Support promotion of university's Faculty of Applied Sciences program by engaging with prospective students through outreach communication and information sessions, helping them understand academic programs and campus opportunities.
- Strengthen the FAS student community by interacting with current students at campus events and providing peer guidance on navigating academic and campus resources via sharing own academic personal journey.

New Student Orientations Volunteer, Langara College

April 2023 – May 2024

- Welcomed and guided over 500 new students during orientation sessions, ensuring a smooth transition to college life through campus tours, resource explanations, and personalized support.
- Fostered a friendly and inclusive environment by answering questions about academic programs, student services, and campus involvement opportunities, helping newcomers feel confident and connected.

School Captain | Leadership Involvement , PKS International School, India

April 2016 – May 2019

- Led a school House team for 2 consecutive years, organizing and training students for 10+ extracurricular activities including public speaking and games, contributing to winning the *Best House Trophy* both years.
- Served as class monitor for 8 years by monitoring class for disruptions in teacher's absence or leave and supporting teachers in test marking which maintained class discipline; anchored school's Annual Event 4 times